

Letters to the Editor

Proton Beam Therapy for Uveal Melanoma

Dear Editor:

Charged-particle irradiation is a relatively new therapy for uveal melanoma, particularly when compared with enucleation. As with any technologic development, the relative value of this technique can only be determined through long and careful study and statistical evaluation of the patients involved. Two articles related to proton beam irradiation were recently published; one of Kincaid et al (*Ophthalmology* 1988; 95:982-91) and the second by Gragoudas et al (*Ophthalmology* 1988; 95:992-9). I have had the opportunity to examine an enucleated globe from a patient whose choroidal melanoma had been treated with proton beam and I would like to add it to the five discussed by Dr. Kincaid et al.

The patient involved is a 72-year-old man diagnosed as having a choroidal malignant melanoma of the right eye in 1984. After the placement of tantalum clips, he received proton beam irradiation for therapy. Subsequently, the lesion remained stable in size and no distant metastases could be found. A cataract recently developed and, because there was evidence of corneal disease as well, he underwent a combination penetrating keratoplasty, extracapsular cataract extraction, and posterior chamber intraocular lens insertion. After surgery, he suffered a sterile corneal melt with imminent perforation. The eye was enucleated in June 1988. Transillumination of the enucleated globe showed a 1.2-cm defect laterally arising 0.7 cm posterior to the limbus. Four tantalum clips formed a diamond pattern over the defect. Sectioning of the globe demonstrated a 1.1-cm diameter, 0.6-cm thick moderately pigmented lesion arising in the choroid. The intraocular lens was in place and the cornea markedly thinned. Results of histologic examination of the tumor showed a spindle-B melanoma with large areas of cellular preservation. One mitotic figure could be identified per 40 high-power fields. Superficial invasion of the underlying sclera was present with evidence of radiation necrosis of collagen within the sclera.

This patient adds a sixth case to the series reported by Kincaid and associates indicating that viable tumor cells may remain in a choroidal melanoma after proton beam irradiation. Gragoudas and co-workers are to be commended for their efforts in the evaluation of this technique, but despite their recent article, the doubt still must remain as to the relative efficacy of proton beam therapy until patients have been followed for a longer period of time. Because one of the aims of proton beam therapy is to preserve useful vision, it is quite likely that more patients will undergo cataract extraction or penetrating keratoplasty on an eye containing an irradiated melanoma which may contain viable tumor cells despite demonstrating current clinical criteria of a treatment success, resulting in an increased risk of metastasis. An immaculately con-

trolled cohort of treated patients must be followed to determine the effectiveness of the therapy and what findings a practicing ophthalmologist may use to ascertain the sterility of a tumor. As a pathologist involved in the Collaborative Ocular Melanoma Study (COMS) discussed by Straatsma et al in the same issue (*Ophthalmology* 1988; 95:1000-4), I have the privilege of participating in such a well-controlled study. Perhaps an arrangement could be worked out whereby the results obtained by Gragoudas et al and those found in the COMS could be compared statistically. Until that time, I seriously doubt whether the preference by a physician for one mode of therapy over another can be made at a level above that of simple personal prejudice.

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Authors' Reply

Dear Editor:

We appreciate Dr. Lawton's interest in our article. His main assumption based on the histopathology of an enucleated eye after proton irradiation is the same as the one discussed in the article by Kincaid et al. These authors assume that the presence of morphologically intact cells in a tissue after irradiation is evidence of remaining viable cells, which have the capability of recurrent growth and metastasis. However, viability from the radiobiologic point of view should be defined only as the capacity of a cell for sustained proliferation.

Early work by Suit and Gallager¹ has demonstrated that above a certain dose, irradiated tumor cells, which appear viable by morphologic criteria are incapable of sustained proliferation. It also has been shown that nonviable cells according to the above definition may complete a few cell division cycles before cell lysis.² Thus, the presence of some mitotic figures in irradiated melanomas does not indicate that irradiation has failed to sterilize these tumors. We and others have published histopathology reports of eyes enucleated after charged particle therapy. Although these findings provide some information regarding changes of irradiated tumors, the presence of morphologically intact cells is not evidence that these cells are viable.

As Green wrote,³ in the same issue, "A cell which has received a lethal dose of radiation (which is more than 6000 rad) may appear histologically and ultrastructurally unremarkable although it is incapable of cellular division, and hence cannot establish a metastasis."

The only proof of cell viability after irradiation is local recurrence of the treated tumor. In our series of more than 1000 patients treated, the local recurrence rate is less than 1.5% which is quite remarkable when compared with